

Decimal to hexadecimal conversion

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Note that **r** will be in the range 0 to 15 (in hexadecimal, 0 to F)

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3. If ***q*** is zero, stop; otherwise set ***n* = *q*** and go back to step 1

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Example: we'll convert decimal **6823** to hexadecimal, step by step

Decimal to hexadecimal conversion

Convert decimal **6823** to hexadecimal, step by step:

Step	Division	Quotient	Remainder
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Decimal to hexadecimal conversion

Convert decimal **6823** to hexadecimal, step by step:

Step	Division	Quotient	Remainder
1	6823/16		

Decimal to hexadecimal conversion

Convert decimal **6823** to hexadecimal, step by step:

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1	6823/16	426	7

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Step	Division	Quotient	Remainder
1	6823/16	426	7



7



7

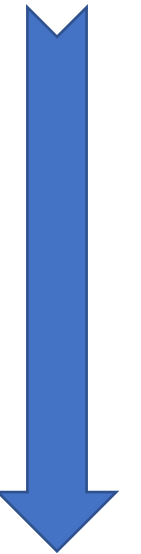
Decimal to hexadecimal conversion

Convert decimal **6823** to hexadecimal, step by step:

Step	Division	Quotient	Remainder
1	6823/16	426	7
2	426/16		



7



7

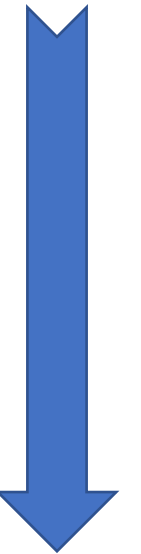
Decimal to hexadecimal conversion

Convert decimal **6823** to hexadecimal, step by step:

Step	Division	Quotient	Remainder
1	6823/16	426	7
2	426/16	26	10



7



7

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1	6823/16	426	7
2	426/16	26	10

7

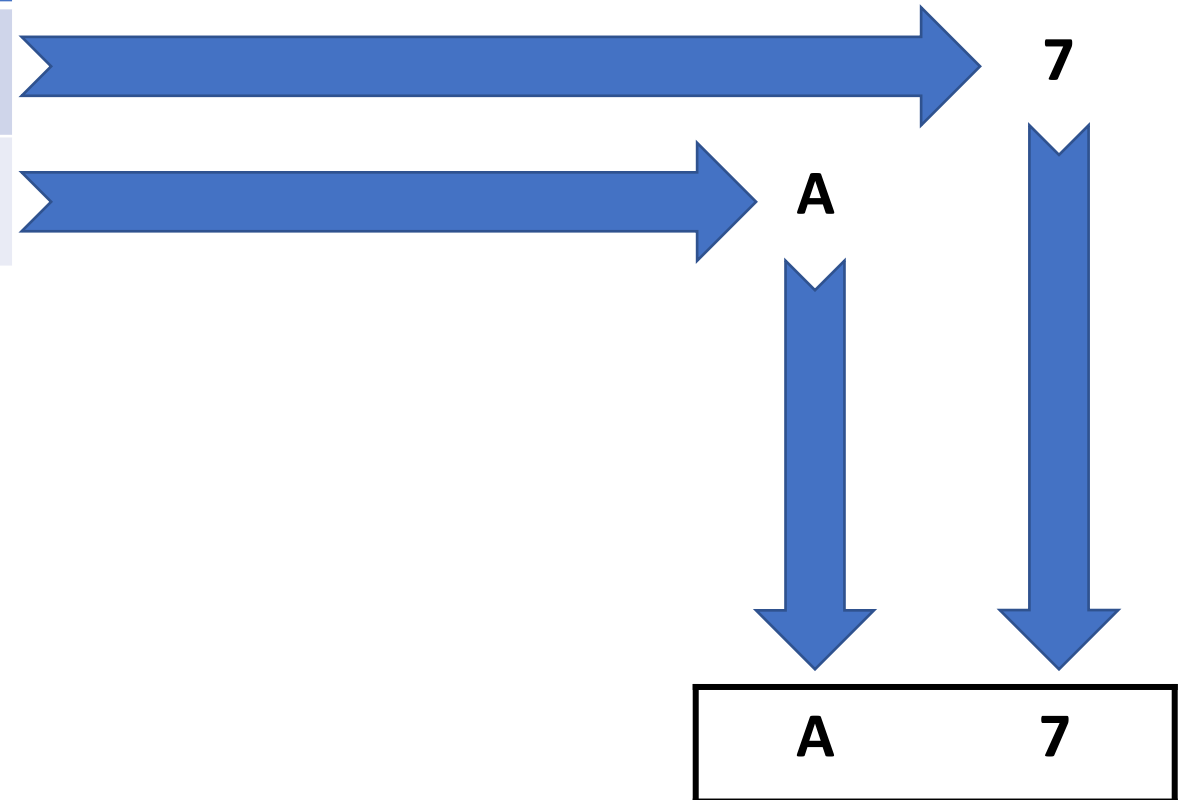
A

7

Decimal to hexadecimal conversion

Convert decimal **6823** to hexadecimal, step by step:

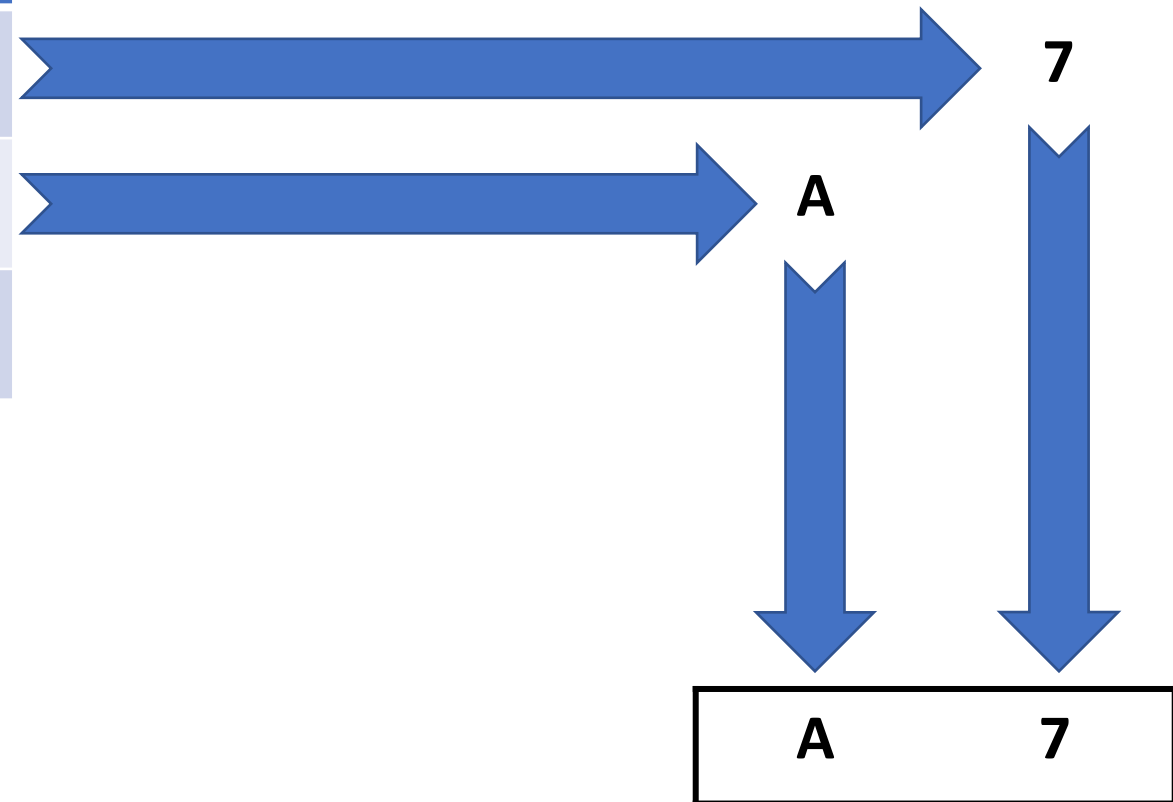
Step	Division	Quotient	Remainder
1	6823/16	426	7
2	426/16	26	10



Decimal to hexadecimal conversion

Convert decimal **6823** to hexadecimal, step by step:

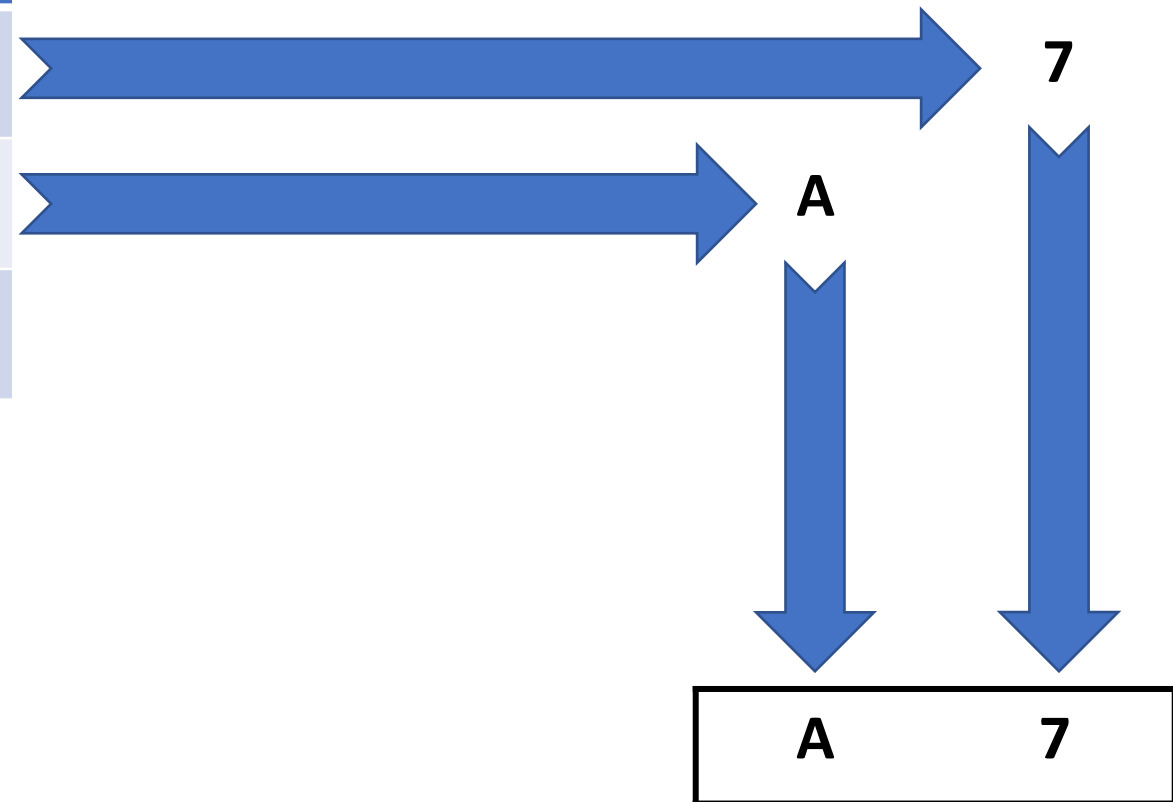
Step	Division	Quotient	Remainder
1	6823/16	426	7
2	426/16	26	10
3	26/16		



Decimal to hexadecimal conversion

Convert decimal **6823** to hexadecimal, step by step:

Step	Division	Quotient	Remainder
1	6823/16	426	7
2	426/16	26	10
3	26/16	1	10



Decimal to hexadecimal conversion

Convert decimal **6823** to hexadecimal, step by step:

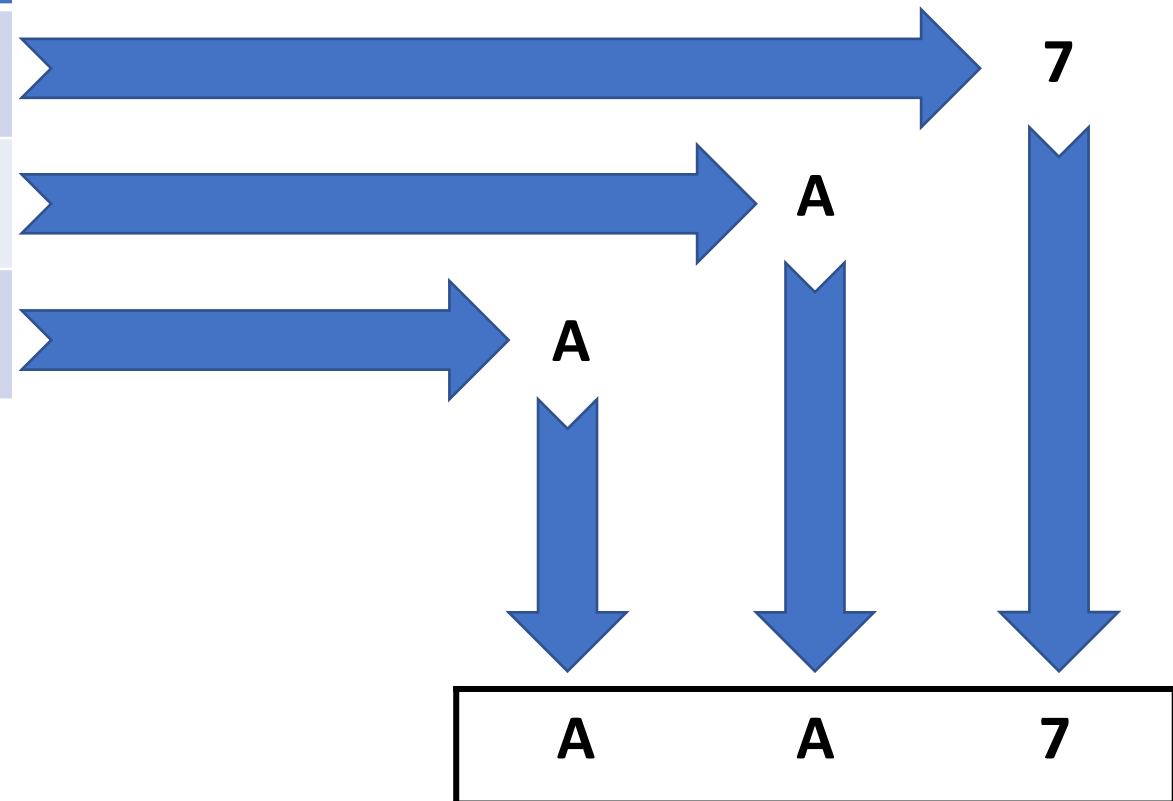
Step	Division	Quotient	Remainder
1	6823/16	426	7
2	426/16	26	10
3	26/16	1	10

The diagram illustrates the conversion process. Three horizontal blue arrows point from the remainder columns of the table to the right. The first arrow points to the digit **7**. The second and third arrows point to the digit **A**. From these **A** digits, two vertical blue arrows point down to a final box containing **A** and **7** in that order from top to bottom.

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3	26/16	1	10



Decimal to hexadecimal conversion

Convert decimal **6823** to hexadecimal, step by step:

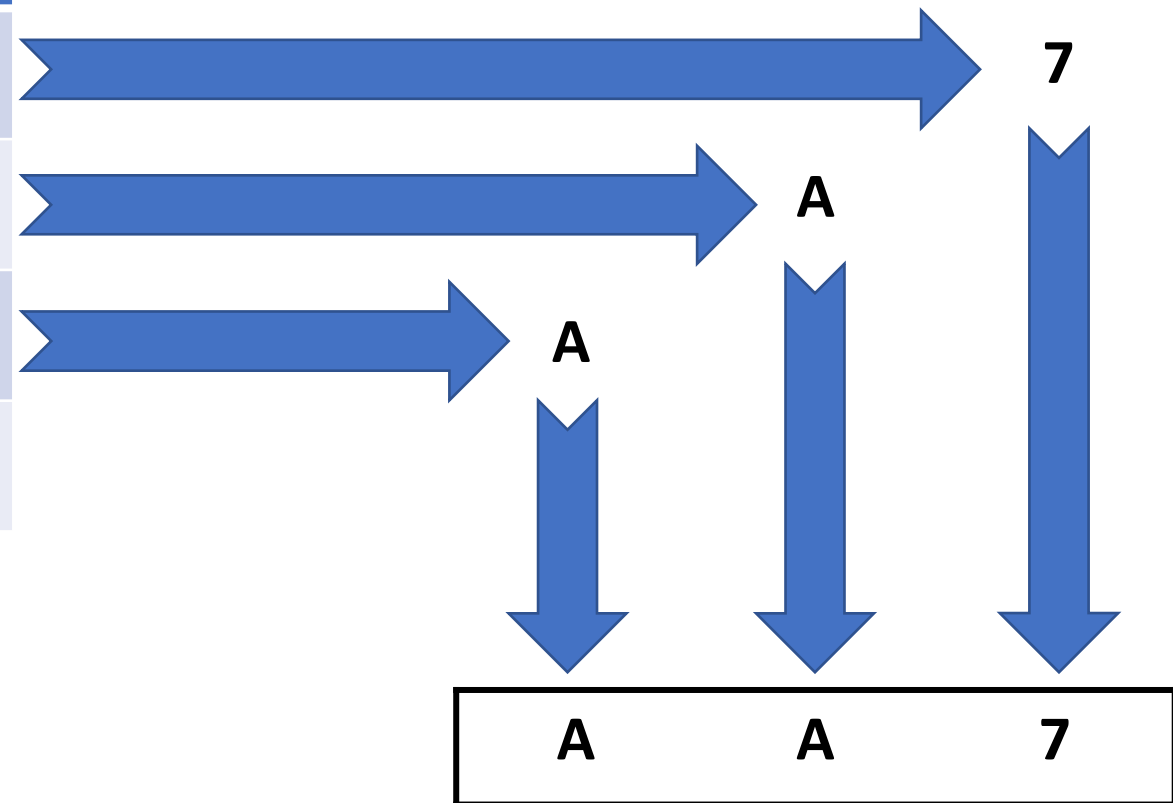
Step	Division	Quotient	Remainder
1	6823/16	426	7
2	426/16	26	10
3	26/16	1	10
4	1/16		

The diagram illustrates the conversion of remainders to hexadecimal digits. Blue arrows point from the remainder column of the table to the digits 7, A, and A. These digits are then placed into a box at the bottom, representing the final hexadecimal result 7AA.

Decimal to hexadecimal conversion

Convert decimal **6823** to hexadecimal, step by step:

Step	Division	Quotient	Remainder
1	6823/16	426	7
2	426/16	26	10
3	26/16	1	10
4	1/16	0	1



Decimal to hexadecimal conversion

Convert decimal **6823** to hexadecimal, step by step:

Step	Division	Quotient	Remainder
1	6823/16	426	7
2	426/16	26	10
3	26/16	1	10
4	1/16	0	1

The diagram illustrates the conversion of remainders to hexadecimal digits and their assembly into the final result. Blue arrows point from the remainders in the table to their corresponding hexadecimal digits: 7, A, A, and 1. These digits are then arranged in a box to form the final hexadecimal value 7AA1.

7

A

A

1

A

A

7

Decimal to hexadecimal conversion

Convert decimal **6823** to hexadecimal, step by step:

Step	Division	Quotient	Remainder
1	6823/16	426	7
2	426/16	26	10
3	26/16	1	10
4	1/16	0	1

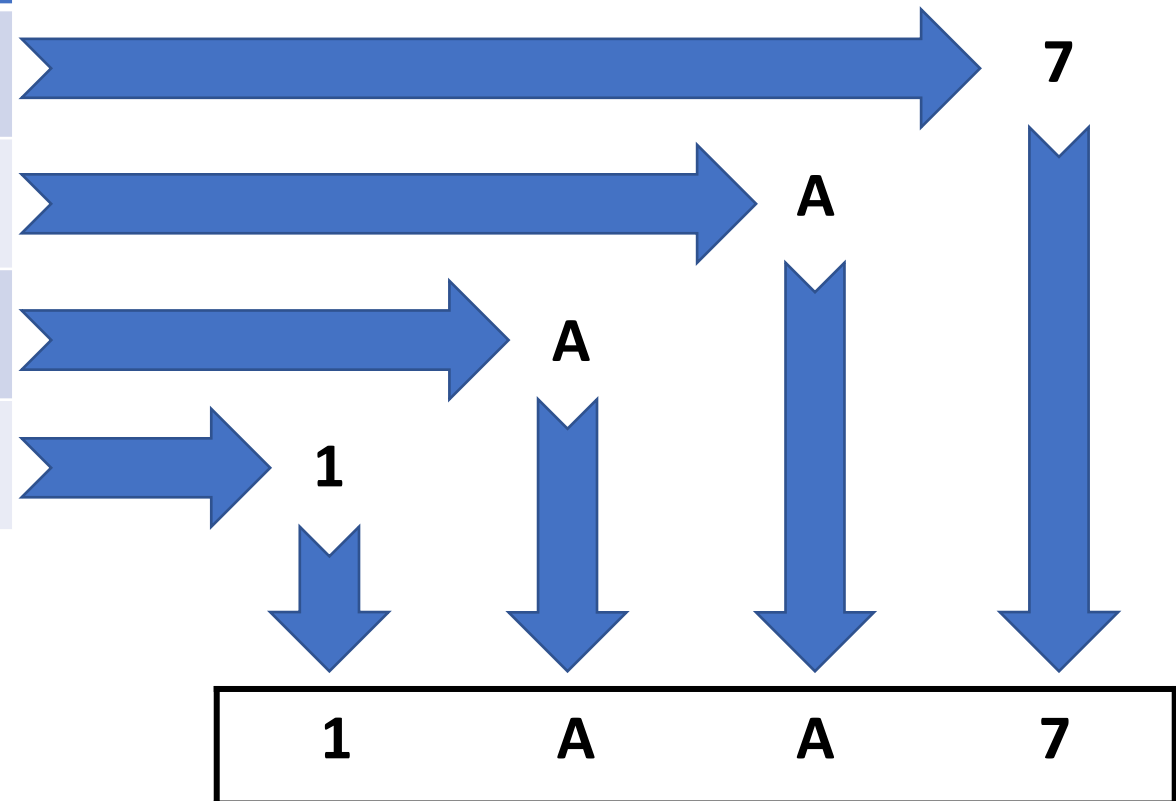
1 A A 7

Decimal to hexadecimal conversion

Convert decimal **6823** to hexadecimal, step by step:

Step	Division	Quotient	Remainder
1	6823/16	426	7
2	426/16	26	10
3	26/16	1	10
4	1/16	0	1

Stop!



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Convert decimal **6823** to hexadecimal, step by step:

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1	6823/16	426	7
2	426/16	26	10
3	26/16	1	10
4	1/16	0	1

Stop!

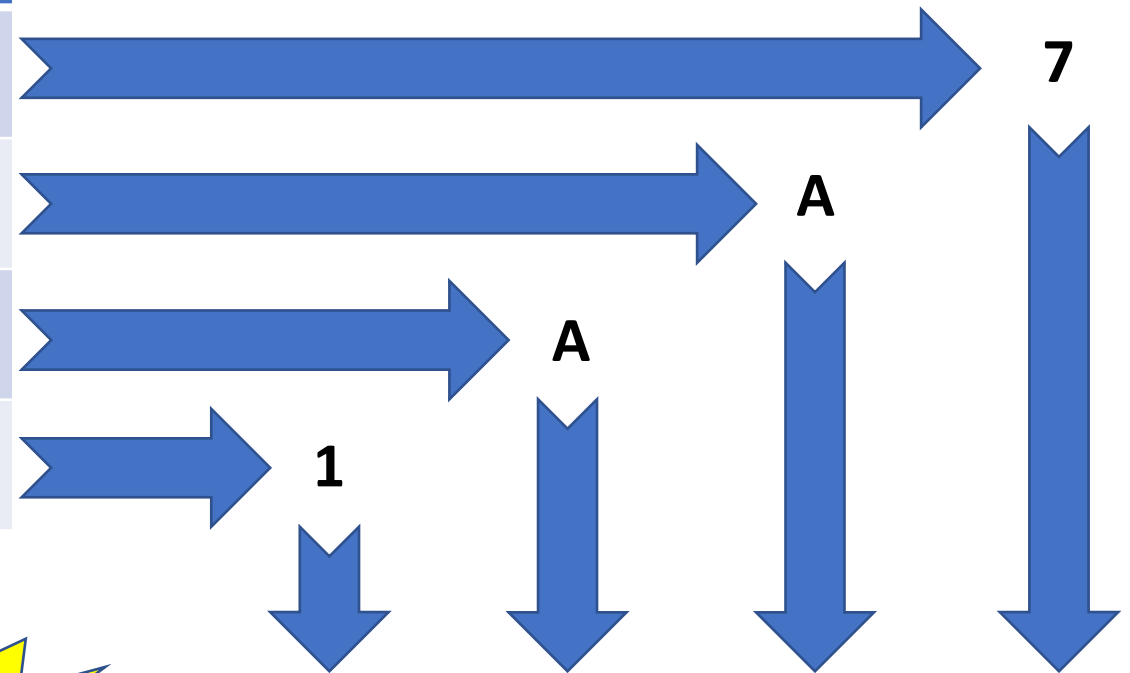
Result:

1

A

A

7



Decimal to binary conversion

- Converting from decimal to binary can take many steps, especially for large numbers
- The easiest way to convert decimal to binary is to convert to hexadecimal first, then replace each digit
- For example, after converting decimal **6823** to hex **1AA7**, going from that to binary is easy:

Hexadecimal...	1	A	A	7
...is binary:	1	1010	1010	0111